

Supplementary Materials A: Generalization

To generalize the analysis, we consider the following setup. Here, $v_{H;i}$ and $v_{L;i}$ are drawn from the same log-concave distribution family. Their supports are still $[\beta_H, \beta_H + 1]$ and $[\beta_L, \beta_L + 1]$, and the cumulative distribution functions are given by $\Psi[v + \gamma(p_H - p_L) - \beta_H]$ and $\Psi(v - \beta_L)$.^{S1} $v_{S;i}$ follows a log-concave distribution with the support $[\theta, \theta + 1]$ and the baseline cumulative distribution function $K(v - \theta)$. $v_{H;i}$, $v_{L;i}$, and $v_{S;i}$ are mutually independent. $\Psi(\cdot)$ and $K(\cdot)$ are twice continuously differentiable over $[0, 1]$, with $\Psi(0) = K(0) = 0$ and $\Psi(1) = K(1) = 1$. The probability density functions $\psi(\cdot)$ and $\kappa(\cdot)$ are positive over $[0, 1]$.

The interior demand functions are given as

$$D_H(p_H, p_L) = 1 - \Psi[p_H + \gamma(p_H - p_L) - \beta_H],$$

$$D_L(p_L) = 1 - \Psi(p_L - \beta_L),$$

$$N(f) = 1 - K(f - \theta).$$

The profit of the platform is

$$\begin{aligned} \pi(p_H, p_L, f) = & (p_H + f)\{1 - \Psi[p_H + \gamma(p_H - p_L) - \beta_H]\}[1 - K(f - \theta)] + \\ & (p_L + f)[1 - \Psi(p_L - \beta_L)][1 - K(f - \theta)]. \end{aligned}$$

No corner solution. There does not exist corner equilibrium if $\psi(\cdot)$ and $\kappa(\cdot)$ on $[0, 1]$ satisfy

$$\kappa(0) < (1 + \gamma)\psi(0) < \frac{1}{(\beta_H + \theta)}, \quad (\text{S1})$$

$$\frac{1 - K(\frac{1}{(1+\gamma)\psi(0)} - \beta_H - \theta)}{\kappa(\frac{1}{(1+\gamma)\psi(0)} - \beta_H - \theta)} < \frac{1}{(1 + \gamma)\psi(0)} - (\beta_H - \beta_L), \quad (\text{S2})$$

$$\psi(1) > \frac{\gamma}{(1 + \gamma)(1 + \beta_L + \theta)}. \quad (\text{S3})$$

^{S1}In the case of no fairness concerns, the cumulative distribution function of high-type consumers is given by $\Psi(v - \beta_H)$.

Proof. Similarly to the main analysis, since there is no profit for $N = 0$, any deviation from $N = 0$ to a positive N should be profitable. Hence, $N = 0$ cannot be in equilibrium.

Given $N > 0$ (and $N \leq 1$ as in the setup), we first consider cases where high-type consumers are unsupported. The first-order derivative of the profit with respect to p_H at $D_H = 0$:

$$\left. \frac{\partial \Pi}{\partial p_H} \right|_{D_H=0} = -(1 + \gamma)(p_H + f)\psi[p_H + \gamma(p_H - p_L) - \beta_H]N < 0,$$

since all the factors are positive. This implies that $D_H = 0$ cannot be in equilibrium.

Next, we consider cases where sellers are fully supported. The first-order derivative of the profit with respect to f at $N = 1$ is given as

$$\left. \frac{\partial \Pi}{\partial f} \right|_{N=1} = D_H + D_L - \kappa(0)[(p_H + \theta)D_H + (p_L + \theta)D_L].$$

For $p_H \geq p_L$, we know

$$\left. \frac{\partial \Pi}{\partial f} \right|_{N=1} \geq (D_H + D_L)[1 - \kappa(0)(p_H + \theta)].$$

If D_H is interior, by the FOC with respect to p_H , we get

$$p_H + \theta \leq p_H + f = \frac{1}{1 + \gamma} \frac{1 - \Psi(p_H + \gamma(p_H - p_L) - \beta_H)}{\psi(p_H + \gamma(p_H - p_L) - \beta_H)}.$$

For the log-concave distribution, $\frac{1 - \Psi(x)}{\psi(x)}$ decreases in x , implying that

$$p_H + \theta \leq \frac{1}{1 + \gamma} \frac{1 - \Psi(p_H + \gamma(p_H - p_L) - \beta_H)}{\psi(p_H + \gamma(p_H - p_L) - \beta_H)} \leq \frac{1}{1 + \gamma} \frac{1 - \Psi(0)}{\psi(0)} = \frac{1}{(1 + \gamma)\psi(0)}.$$

Since $D_H + D_L > 0$ together with (S1), we get

$$\left. \frac{\partial \Pi}{\partial f} \right|_{N=1} \geq (D_H + D_L)[1 - \kappa(0)(p_H + \theta)] \geq (D_H + D_L)\left[1 - \frac{\kappa(0)}{(1 + \gamma)\psi(0)}\right] > 0.$$

If $D_H = 1$,

$$\left. \frac{\partial \Pi}{\partial f} \right|_{N=1, D_H=1} \geq (D_H + D_L)[1 - \kappa(0)(\beta_H + \theta)] > (D_H + D_L)[1 - \psi(0)(1 + \gamma)(\beta_H + \theta)] > 0.$$

In summary, given that $p_H < p_L$ cannot be in equilibrium,^{S2} $N = 1$ cannot be in equilibrium.

Then, we consider cases where high-type consumers are fully supported. We know that $p_H = p_H^{\min} = \frac{\beta_H + \gamma p_L}{1 + \gamma}$ for $D_H = 1$. Therefore, we have

$$\left. \frac{\partial \Pi}{\partial p_H} \right|_{D_H=1} = \{1 - [\beta_H + \gamma p_L + (1 + \gamma)f]\psi(0)\}N.$$

For interior N , the FOC with respect to f requires

$$[1 - K(f - \theta)](D_H + D_L) - \kappa(f - \theta)[(p_H + \theta)D_H + (p_L + \theta)D_L] = 0.$$

This implies

$$\frac{1 - K(f - \theta)}{\kappa(f - \theta)} = \frac{(p_H + \theta)D_H + (p_L + \theta)D_L}{D_H + D_L} \geq p_L + f \geq \beta_L + f.$$

Now, for f that satisfies the above inequality, we define

$$\Phi(f) = f + \beta_L - \frac{1 - K(f - \theta)}{\kappa(f - \theta)} \leq 0.$$

By the mean value theorem, there exists f^{mid} between f and $\frac{1}{(1 + \gamma)\psi(0)} - \beta_H$ such that

$$\Phi(f) - \Phi\left[\frac{1}{(1 + \gamma)\psi(0)} - \beta_H\right] = \Phi'(f^{\text{mid}})\left[f - \frac{1}{(1 + \gamma)\psi(0)} + \beta_H\right].$$

Here, by (S2) together with $\Phi(f) < 0$, we know the left side is non-positive. Since $\kappa(\cdot)$ is

^{S2}We omit the detailed proof here. Given the log-concave distributions and the fairness concern term, it is easy to find this result.

log-concave, $\Phi'(f^{\text{mid}}) > 0$. Therefore, we get

$$f \leq \frac{1}{(1+\gamma)\psi(0)} - \beta_H.$$

In summary, for f that satisfies the above inequality and $p = p^{\min}$, we have

$$\beta_H + \gamma p_L + (1+\gamma)f \leq (1+\gamma)\beta_H + (1+\gamma)\left[\frac{1}{(1+\gamma)\psi(0)} - \beta_H\right] = \frac{1}{\psi(0)},$$

implying $\frac{\partial \Pi}{\partial p_H} \Big|_{D_H=1} > 0$. Therefore, $D_H = 1$ cannot be in equilibrium.

Finally, we consider the cases where low-type consumers are unsupported and where they are fully supported. The first-order derivative of the profit with respect to p_L at $D_L = 0$ is

$$\frac{\partial \Pi}{\partial p_L} \Big|_{D_L=0} = \{\gamma(p_H + f)\psi[p_H + \gamma(p_H - 1 - \beta_L) - \beta_H] - (1 + \beta_L + f)\psi(1)\}N. \quad (\text{S4})$$

The FOC with respect to p_H implies

$$(p_H + f)\psi[p_H + \gamma(p_H - p_L) - \beta_H] = \frac{1 - \Psi[p_H + \gamma(p_H - p_L) - \beta_H]}{1 + \gamma}. \quad (\text{S5})$$

Plugging (S5) into (S4) together with inequality (S3), we get

$$\frac{\partial \Pi}{\partial p_L} \Big|_{D_L=0} < \frac{\gamma}{1 + \gamma} - (1 + \beta_L + f)\psi(1) < 0,$$

implying that $D_L = 0$ cannot be in equilibrium. For $D_L = 1$ with interior p_H and f ,

$$\begin{aligned} \frac{\partial \Pi}{\partial p_L} \Big|_{D_L=1} &= \{1 + \gamma(p_H + f)\psi[p_H + \gamma(p_H - \beta_L) - \beta_H] - (\beta_L + f)\psi(0)\}N \\ &> \{1 - [\beta_H + \gamma p_L + (1 + \gamma)f]\psi(0)\}N = \frac{\partial \Pi}{\partial p_H} \Big|_{D_H=1} > 0, \end{aligned}$$

implying that $D_L = 1$ cannot be in equilibrium.

Now, we show that the key results continue to hold for interior pricing strategies if the density of consumer distribution varies sufficiently slowly such that

$$\left| \frac{\psi'(\cdot)}{\psi(\cdot)} \right| \leq \varepsilon, \quad (\text{FC})$$

with

$$0 < \varepsilon < \frac{-(1 + 2\gamma^2) + \sqrt{(1 + 2\gamma^2)^2 + 8(2 + 2\gamma - \gamma^2)}}{4}. \quad (\text{SI})$$

Interior equilibrium. There exists an interior equilibrium.

Proof. We hereafter denote

$$\Psi_H = \Psi(x_H) = \Psi[p_H + \gamma(p_H - p_L) - \beta_H],$$

$$\Psi_L = \Psi(x_L) = \Psi(p_L - \beta_L),$$

$$K = K(y) = K(p_L - \beta_L),$$

$$\psi_H = \psi(x_H), \quad \psi_L = \psi(x_L), \quad \kappa = \kappa(y),$$

$$\psi'_H = \frac{\partial \psi(x_H)}{\partial x_H}, \quad \psi'_L = \frac{\partial \psi(x_L)}{\partial x_L}, \quad \kappa' = \frac{\partial \kappa(y)}{\partial y}.$$

Accordingly, the FOCs for profit maximization can be written as

$$\frac{\partial \pi}{\partial p_H} = N[1 - \Psi_H - (p_H + f)(1 + \gamma)\psi_H] = 0, \quad (\text{S6})$$

$$\frac{\partial \pi}{\partial p_L} = N[\gamma(p_H + f)\psi_H + 1 - \Psi_L - (p_L + f)\psi_L] = 0, \quad (\text{S7})$$

$$\frac{\partial \pi}{\partial f} = (1 - K)(D_H + D_L) - \kappa[(p_H + f)D_H + (p_L + f)D_L] = 0. \quad (\text{S8})$$

The Hessian matrix of the profit function is given as

$$\begin{bmatrix} \pi_{11} & \pi_{12} & \pi_{13} \\ \pi_{21} & \pi_{22} & \pi_{23} \\ \pi_{31} & \pi_{32} & \pi_{33} \end{bmatrix},$$

where

$$\pi_{11} = \frac{\partial^2 \pi}{\partial p_H^2} = N[-2(1 + \gamma)\psi_H - (p_H + f)(1 + \gamma)^2\psi'_H],$$

$$\pi_{12} = \pi_{21} = \frac{\partial^2 \pi}{\partial p_H \partial p_L} = N[\gamma\psi_H + \gamma(p_H + f)(1 + \gamma)\psi'_H],$$

$$\pi_{22} = \frac{\partial^2 \pi}{\partial p_L^2} = N[-2\psi_L - (p_L + f)\psi'_L - \gamma^2(p_H + f)\psi'_H],$$

$$\pi_{13} = \pi_{31} = \frac{\partial^2 \pi}{\partial p_H \partial f} = N[-(1 + \gamma)\psi_H],$$

$$\pi_{23} = \pi_{32} = \frac{\partial^2 \pi}{\partial p_L \partial f} = N[\gamma\psi_H - \psi_L],$$

$$\pi_{33} = \frac{\partial^2 \pi}{\partial f^2} = -\kappa(D_H + D_L)[2 + \frac{(1 - K)\kappa'}{\kappa^2}].$$

Given inequality [\(FC\)](#), the first leading principle minor:

$$\pi_{11} = N[-2(1 + \gamma)\psi_H - (p_H + f)(1 + \gamma)^2\psi'_H] < N[-2(1 + \gamma)\psi_H + (1 + \gamma)^2\psi_H] < 0.$$

The second leading principal minor:

$$\pi_{11}\pi_{22} - \pi_{12}^2 = N^2\Delta,$$

where

$$\frac{\Delta}{\psi_H\psi_L} = (1 + \gamma)[2 + (1 + \gamma)(p_H + f)\frac{\psi'_H}{\psi_H}][2 + (p_L + f)\frac{\psi'_L}{\psi_L}] - \gamma^2\frac{\psi_H}{\psi_L}.$$

Therefore,

$$N^2 \Delta > 0 \Leftrightarrow (1 + \gamma)[2 + (1 + \gamma)(p_H + f) \frac{\psi'_H}{\psi_H}][2 + (p_L + f) \frac{\psi'_L}{\psi_L}] > \gamma^2 \frac{\psi_H}{\psi_L}. \quad (\text{S9})$$

Since $\psi(x) > 0$ for $x \in [0, 1]$ together with **(FC)**, $|\ln \psi(x)|' = |\psi'(x)/\psi(x)| < \varepsilon$. By the mean value theorem applied to $\ln \psi$, for any $x_1, x_2 \in [0, 1]$, there exists $\bar{x}$ between x_1 and x_2 such that $\ln \psi(x_1) - \ln \psi(x_2) = [\psi'(\bar{x})/\psi(\bar{x})](x_1 - x_2)$. Then, we get

$$|\ln \psi_H - \ln \psi_L| \leq \varepsilon |x_H - x_L| \leq \varepsilon. \quad (\text{S10})$$

Since $\int_0^1 \psi(t) dt = 1$, there exists $x_0 \in [0, 1]$ such that $\psi(x_0) = 1$ and $\ln \psi(x_0) = 0$ by the mean value theorem for integrals. According to **(S10)**, we get for all $x \in [0, 1]$ that

$$\psi(x) = e^{\ln \psi(x) - \ln \psi(x_0)} \in [e^{-\varepsilon}, e^{\varepsilon}]. \quad (\text{S11})$$

According to the first order conditions, $p_L + f \leq p_H + f < \frac{1}{(1+\gamma)\psi(0)}$. For interior p_H, p_L , and f , we have

$$\frac{\psi_H}{\psi_L} \in [e^{-\varepsilon}, e^{\varepsilon}], \quad (\text{S12})$$

$$(p_H + f) \left| \frac{\psi'_H}{\psi_H} \right| < \frac{\varepsilon}{(1 + \gamma)\psi(0)}, \quad (\text{S13})$$

$$(p_L + f) \left| \frac{\psi'_L}{\psi_L} \right| < \frac{\varepsilon}{(1 + \gamma)\psi(0)}. \quad (\text{S14})$$

Since $\Psi(x)$ is log-concave, $[\ln(1 - \Psi(x))]' \leq 0$, implying $-\psi(x)'(1 - \Psi(x)) - \psi(x)^2 \leq 0$. Therefore, for all $x \in [0, 1]$,

$$\frac{[1 - \Psi(x)]\psi'(x)}{\psi(x)^2} > -1.$$

Given this, for pricing strategies that satisfy the FOC with respect to p_H ,

$$2 + (1 + \gamma)(p_H + f) \frac{\psi'_H}{\psi_H} > 1.$$

Together with (S12), inequality (S9) must be satisfied if

$$2(1 + \gamma) + (p_L + f)(1 + \gamma) \frac{\psi'_L}{\psi_L} \geq \gamma^2 e^\varepsilon.$$

By (S11) and (S12) through (S14), we get

$$2(1 + \gamma) + (p_L + f)(1 + \gamma) \frac{\psi'_L}{\psi_L} \geq 2(1 + \gamma) - \frac{\varepsilon}{\psi(0)} \geq 2(1 + \gamma) - \varepsilon e^\varepsilon. \quad (\text{S15})$$

By (SI), we know $\varepsilon < 1$. Therefore, $e^\varepsilon < 1 + 2\varepsilon$ and $2\varepsilon^2 + (1 + 2\gamma)\varepsilon - (2 + 2\gamma - \gamma^2) < 0$.

Plugging these inequalities into (S15) gives

$$2(1 + \gamma) + (p_L + f)(1 + \gamma) \frac{\psi'_L}{\psi_L} \geq 2(1 + \gamma) - \varepsilon e^\varepsilon > 2(1 + \gamma) - \varepsilon(1 + 2\varepsilon) > \gamma^2(1 + 2\varepsilon) > \gamma^2 \frac{\psi_H}{\psi_L}.$$

Therefore, $N^2\Delta > 0$, implying that the second leading principal minor is positive. Since this ensures that the Jacobian of the consumer-side first-order conditions is non-singular, the optimal prices vary continuously with f , implying π is continuous in f on the compact support $[\theta, \theta + 1]$. Therefore, an internal equilibrium for profit maximization exists by the Weierstrass extreme value theorem.

Trade-off between the consumer-side price difference and the seller-side price.

For $p_H > p_L$, the best responses $p_H^*(f)$, $p_L^*(f)$ for a given f satisfy:

- (i) There exists $\frac{d[p_H^*(f) - p_L^*(f)]}{df} > 0$ for $\gamma = 0$.
- (ii) There exists $\gamma^{\text{bench}} \in (0, 1)$ such that $\frac{d[p_H^*(f) - p_L^*(f)]}{df} < 0$ for all $\gamma > \gamma^{\text{bench}}$.

Intuitively, (i) indicates that a positive relationship between the consumer-side price difference and the price for sellers is possible for profit maximization when consumers have no fairness concerns, while (ii) indicates that the above possibility no longer exists when

consumers have enough strong fairness concerns. These results imply that the presence of fairness concerns induces an extra trade-off between the price difference on the consumer side and the price for sellers.

Proof. Applying the implicit function theorem and totally differentiating the FOCs (S6) and (S7) with respect to f yield

$$\begin{pmatrix} \pi_{11} & \pi_{12} \\ \pi_{12} & \pi_{22} \end{pmatrix} \begin{pmatrix} \frac{dp_H^*(f)}{df} \\ \frac{dp_L^*(f)}{df} \end{pmatrix} + \begin{pmatrix} \pi_{13} \\ \pi_{23} \end{pmatrix} = 0.$$

Solving the system of equations by Cramer's rule gives

$$\begin{cases} \frac{dp_H^*(f)}{df} = -\frac{\pi_{22}\pi_{13} - \pi_{12}\pi_{23}}{\pi_{11}\pi_{22} - \pi_{12}^2} \\ \frac{dp_L^*(f)}{df} = -\frac{\pi_{11}\pi_{23} - \pi_{12}\pi_{13}}{\pi_{11}\pi_{22} - \pi_{12}^2}. \end{cases}$$

Then, we get

$$\frac{d[p_H^*(f) - p_L^*(f)]}{df} = -\frac{(\pi_{22} + \pi_{12})\pi_{13} - (\pi_{11} + \pi_{12})\pi_{23}}{\pi_{11}\pi_{22} - \pi_{12}^2} = -\frac{R}{\Delta},$$

where

$$\frac{R}{\psi_H\psi_L} = \gamma(1 + \frac{\psi_H}{\psi_L}) - (1 + \gamma)[(p_H + f)\frac{\psi'_H}{\psi_H} - (p_L + f)\frac{\psi'_L}{\psi_L}].$$

For $\gamma = 0$,

$$\frac{R}{\psi_H\psi_L} = (p_L + f)\frac{\psi'_L}{\psi_L} - (p_H + f)\frac{\psi'_H}{\psi_H}.$$

Therefore, given $p_H > p_L$, $R < 0$ if $0 < \frac{\psi'_L}{\psi_L} \leq \frac{\psi'_H}{\psi_H}$ (e.g., truncated exponential distribution with a strictly increasing probability density function). In this case,

$$\frac{d[p_H^*(f) - p_L^*(f)]}{df} = -\frac{R}{\Delta} > 0.$$

For $\gamma \in (0, 1)$, by (S12) through (S14), we get

$$\frac{R}{\psi_H \psi_L} > \gamma[1 + e^{-\varepsilon}] - \frac{2\varepsilon}{\psi(0)}.$$

There exists

$$\gamma^{\text{bench}} = \frac{2\varepsilon}{[1 + e^{-\varepsilon}]\psi(0)}, \quad (\text{S16})$$

for all $\gamma > \gamma^{\text{bench}}$

$$\frac{d[p_H^*(f) - p_L^*(f)]}{df} = -\frac{R}{\Delta} < 0.$$

In addition, since $\frac{2\varepsilon}{[1+e^{-\varepsilon}]\psi(0)}$ in (S16) is increasing in ε , a smaller ε lowers the threshold γ^{bench} , making it easier for the price gap to decrease in f .

Existence of uniform pricing in equilibrium. Here, we have the following results:

- (i) Interior equilibrium prices must satisfy $p_H^* > p_L^*$ when $\gamma = 0$.
- (ii) There exists $\beta_H, \beta_L, \gamma, \psi(\cdot)$, and $\kappa(\cdot)$ such that an interior equilibrium with $p_H^* = p_L^*$ exists for $\theta > \theta^{\text{bench}}$.

Proof of (i). Assume $p_H^* \leq p_L^*$ when $\gamma = 0$. The FOC with respect to p_i becomes

$$\frac{1 - \Psi(p_i - \beta_i)}{\psi(p_i - \beta_i)} = p_i + f.$$

Since $p_H^* \leq p_L^*$ and $\beta_H > \beta_L$, we have $p_H^* - \beta_H < p_L^* - \beta_L$. Because the distribution is log-concave, $\frac{1-\Psi(x)}{\psi(x)}$ is decreasing in x . Therefore,

$$\frac{1 - \Psi(p_H^* - \beta_H)}{\psi(p_H^* - \beta_H)} > \frac{1 - \Psi(p_L^* - \beta_L)}{\psi(p_L^* - \beta_L)},$$

implying $p_H^* > p_L^*$. This contradicts the assumption $p_H^* \leq p_L^*$. Hence, the platform must price-discriminate among consumers when there are no fairness concerns.

Illustration of (ii). The main analysis has already shown that the platform can adopt uniform pricing in equilibrium. Here, we present two specific cases with simulated $p_H^* - p_L^*$ and f^* with respect to θ . In both cases, when consumers have fairness concerns, the consumer-side price difference decreases in the seller-side price and maintains uniform pricing for large enough f^* . By comparison, the consumer-side price difference is always positive when consumers do not have fairness concerns.

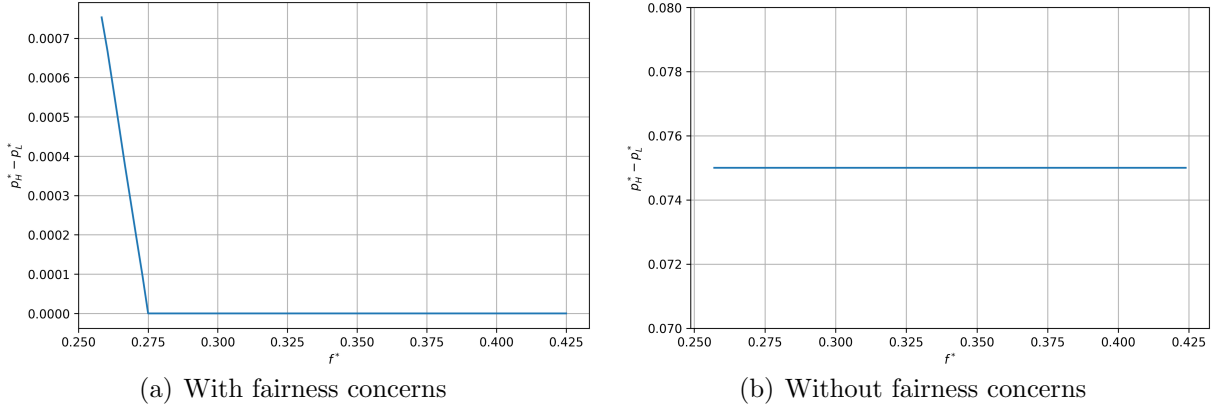


Figure S1: Price difference by γ . Distribution: $v_{H;i} \sim U(\beta_H, \beta_H + 1)$, $v_{L;i} \sim U(\beta_L, \beta_L + 1)$, and $v_{S;i} \sim U(\theta, \theta + 1)$. Parameter values: $\beta_H = 0.3$, $\beta_L = 0.15$, $\theta \in (0, 0.25)$, and $\gamma = 0.1$.

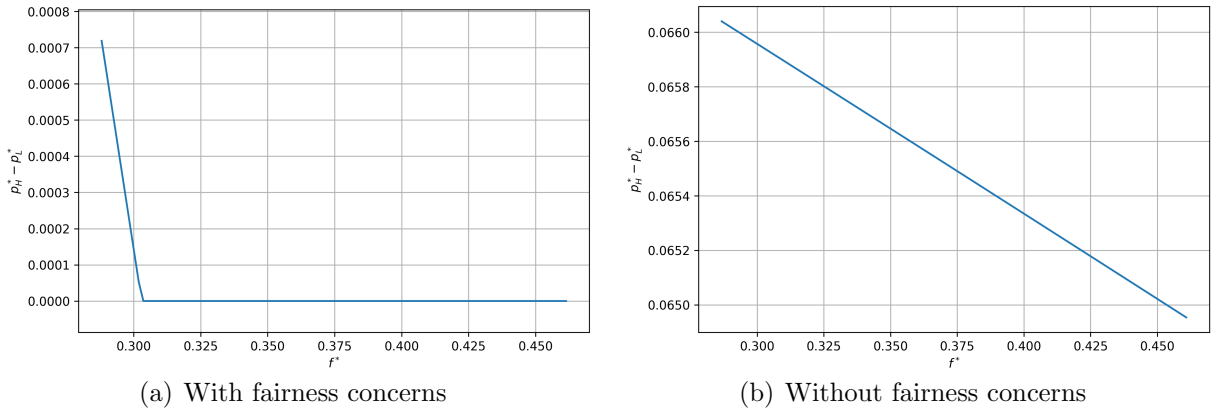


Figure S2: Consumer-side price difference and seller-side price in equilibrium. Distribution: $v_{H;i} \sim \text{TrunExp}(\lambda, \beta_H, \beta_H + 1)$, $v_{L;i} \sim \text{TrunExp}(\lambda, \beta_L, \beta_L + 1)$, and $v_{S;i} \sim U(\theta, \theta + 1)$. Parameter values: $\lambda = 0.3$, $\beta_H = 0.3$, $\beta_L = 0.15$, $\theta \in (0, 0.25)$, and $\gamma = 0.08$.

Supplementary Materials B: Other Omitted Proofs

B1 Proof of Proposition 6

With the generalized form of consumers' psychological responses to price differences, the platform's total profit is

$$\Pi^G = p_H D_H^G(p_H, p_L) N(f) + p_L D_L^G(p_H, p_L) N(f).$$

The first-order conditions for profit maximization with respect to p_H and p_L are

$$\frac{\partial \Pi^G}{\partial p_H} = 0 \Leftrightarrow D_H^G + (p_H + f) \frac{\partial D_H^G}{\partial p_H} + (p_L + f) \frac{\partial D_L^G}{\partial p_H} = 0, \quad (\text{S17})$$

$$\frac{\partial \Pi^G}{\partial p_L} = 0 \Leftrightarrow (p_H + f) \frac{\partial D_H^G}{\partial p_L} + D_L^G + (p_L + f) \frac{\partial D_L^G}{\partial p_L} = 0. \quad (\text{S18})$$

Solving the simultaneous equations gives the optimal consumer-side prices in response to a given price for sellers:

$$p_H^*(f) = \frac{(2 + 2\eta)\beta_H + (\eta + \gamma)(1 + \beta_L)}{4 + 4\gamma + 4\eta - 2\gamma\eta - \gamma^2 - \eta^2} - \frac{2 + 3\gamma + \eta + 2\gamma\eta - \gamma^2 - \eta^2}{4 + 4\gamma + 4\eta - 2\gamma\eta - \gamma^2 - \eta^2} f, \quad (\text{S19})$$

$$p_L^*(f) = \frac{(\eta + \gamma)\beta_H + (2 + 2\gamma)(1 + \beta_L)}{4 + 4\gamma + 4\eta - 2\gamma\eta - \gamma^2 - \eta^2} - \frac{2 + \gamma + 3\eta + 2\gamma\eta - \gamma^2 - \eta^2}{4 + 4\gamma + 4\eta - 2\gamma\eta - \gamma^2 - \eta^2} f. \quad (\text{S20})$$

Thus, the equilibrium prices such that $p_H > p_L$ satisfy

$$p_H^* - p_L^* = B^G + T^G f^*,$$

with $B^G = \frac{(2+\eta-\gamma)(1+\beta_H)-(2-\eta+\gamma)(1+\beta_L)}{4+4\gamma+4\eta-2\gamma\eta-\gamma^2-\eta^2}$, and $T^G = -\frac{2\gamma-2\eta}{4+4\gamma+4\eta-2\gamma\eta-\gamma^2-\eta^2}$.

Here, given $\gamma > \eta > 0$, we have $B^G > B > 0$ and

$$\frac{\partial T^G}{\partial \eta} = \frac{8 + 16\eta - 4\gamma\eta + 2\gamma^2 - 6\eta^2}{(4 + 4\gamma + 4\eta - 2\gamma\eta - \gamma^2 - \eta^2)^2} > 0.$$

Consider that $T^G = T$ at $\eta = 0$. We further have $T < T^G < 0$.

B2 Proof of Proposition 7

When the proportions of high-type and low-type consumers can be different, the platform's total profit becomes

$$\tilde{\Pi} = \frac{2\mu}{1+\mu}(p_H + f)\widetilde{D_H}(p_H, p_L)N(f) + \frac{2}{1+\mu}(p_L + f)D_L(p_L)N(f).$$

For maximizing the platform's profit, the first-order condition with respect to p_H has the same form with equation (6), while the first-order condition with respect to p_L becomes

$$\frac{\partial \tilde{\Pi}}{\partial p_L} = 0 \Leftrightarrow \mu(p_H + f)\frac{\partial D_H}{\partial p_L} + D_L + (p_L + f)\frac{\partial D_L}{\partial p_L} = 0. \quad (\text{S21})$$

Solving the simultaneous equations gives the optimal consumer-side prices in response to a given price for sellers:

$$p_H^*(f) = \frac{2(1 + \beta_H) + \gamma(1 + \beta_L)}{4 + 4\gamma - \mu\gamma^2} - \frac{2 + 3\gamma - \mu\gamma^2}{4 + 4\gamma - \mu\gamma^2}f, \quad (\text{S22})$$

$$p_L^*(f) = \frac{\mu\gamma(1 + \beta_H) - 2(1 + \beta_L)}{4 + 4\gamma - \mu\gamma^2} - \frac{2 + (2 - \mu)\gamma - \mu\gamma^2}{4 + 4\gamma - \mu\gamma^2}f. \quad (\text{S23})$$

Thus, the equilibrium prices such that $p_H^* > p_L^*$ satisfy

$$p_H^* - p_L^* = \tilde{B} + \tilde{T}f^*,$$

with $\tilde{B} = \frac{(2-\mu\gamma)(1+\beta_H)-(2+\gamma)(1+\beta_L)}{4+4\gamma-\mu\gamma^2}$ and $\tilde{T} = -\frac{(1+\mu)\gamma}{4+4\gamma-\mu\gamma^2}$. Here, it is easy to find that $\tilde{B}$ is positive and $\tilde{T}$ is negative as $p_H > p_L$ on the interior, and we also have

$$\frac{\partial \tilde{T}}{\partial \mu} = -\frac{4\gamma + 4\gamma^2 + \gamma^3}{(4 + 4\gamma - \mu\gamma^2)^2} < 0.$$

B3 Discussion on endogenous γ to μ

Here, we further extend the demand of high-type consumers to

$$\widetilde{D}_H(p_H, p_L) = 1 + \beta_H - p_H - \gamma(\mu)(p_H - p_L),$$

where $\mu = \frac{\lambda}{1-\lambda} \in (0, +\infty)$, $\gamma(0) = \gamma_0 \in (0, 1)$, and $\lim_{\mu \rightarrow +\infty} \gamma(\mu) = 0$. We assume $\frac{\partial \gamma(\mu)}{\partial \mu} = 0$ for $\mu < 1$ and $\frac{\partial \gamma(\mu)}{\partial \mu} = \alpha < 0$ for $\mu \geq 1$. This formulates how consumers' aversion to pricing unfairness is endogenous to the proportion of high-type consumers. Recall

$$p_H^* - p_L^* = \widetilde{B} + \widetilde{T}f^*,$$

with $\widetilde{B} = \frac{(2-\mu\gamma)(1+\beta_H)-(2+\gamma)(1+\beta_L)}{4+4\gamma-\mu\gamma^2}$ and $\widetilde{T} = -\frac{(1+\mu)\gamma}{4+4\gamma-\mu\gamma^2}$. In the case of $\gamma(\mu)$, we derive

$$\frac{\partial \widetilde{T}}{\partial \mu} < 0 \Leftrightarrow \frac{\partial \gamma(\mu)}{\partial \mu} > -\frac{\gamma(\mu)(\gamma(\mu) + 2)^2}{(1 + \mu)(\gamma(\mu)^2 \mu + 4)},$$

$$\frac{\partial \widetilde{T}}{\partial \mu} \geq 0 \Leftrightarrow \frac{\partial \gamma(\mu)}{\partial \mu} \leq -\frac{\gamma(\mu)(\gamma(\mu) + 2)^2}{(1 + \mu)(\gamma(\mu)^2 \mu + 4)}.$$

For $\mu < 1$, given that $\frac{\partial \gamma}{\partial \mu} = 0$, there must be

$$\frac{\partial \gamma(\mu)}{\partial \mu} = 0 > -\frac{\gamma(\mu)(\gamma(\mu) + 2)^2}{(1 + \mu)(\gamma(\mu)^2 \mu + 4)}.$$

For $\mu \geq 1$, given that $\frac{\partial \gamma}{\partial \mu} = \alpha < 0$, there must be

$$\frac{\partial \gamma(\mu)}{\partial \mu} = \alpha < \lim_{\mu \rightarrow +\infty} \frac{\gamma(\mu)(\gamma(\mu) + 2)^2}{(1 + \mu)(\gamma(\mu)^2 \mu + 4)} = 0.$$

In this case, it is uncertain how the trade-off between the price difference on the consumer side and the price for sellers changes as μ increases. On the one hand, adding weight to high-type consumers still makes it more profitable to attract high-type consumers relative

to low-type consumers. This strengthens the demand externality from low-type to high-type consumers. On the other hand, in the case where high-type consumers are in the majority (i.e., $\mu \geq 1$), a higher μ additionally implies that consumers' aversion to pricing unfairness gets weaker (i.e., a lower γ). This weakens the demand externality and thus weakens the trade-off.

Here we mark two special cases. First, when high-type consumers are in the minority (i.e., $\mu < 1$), the positive effect of increasing the proportion on the trade-off is definitely greater than the negative effect. Second, when the proportion of high-type consumers is extremely large (i.e., $\mu \rightarrow +\infty$), the positive effect of increasing the proportion on the trade-off is smaller than the negative effect.

B4 Proofs of Lemma 3 and Proposition 8

When the platform can implement seller-side price discrimination, its total profit becomes

$$\begin{aligned}\hat{\pi}(p_H, p_L, f_H, f_L) = & \frac{1}{2}[(p_H + f_H)D_H(p_H, p_L) + (p_L + f_H)D_L(p_L)]N_H(f_H) + \\ & \frac{1}{2}[(p_H + f_L)D_H(p_H, p_L) + (p_L + f_L)D_L(p_L)]N_L(f_L).\end{aligned}$$

The FOCs are

$$\begin{aligned}\frac{\partial \Pi}{\partial p_H} = 0 & \Leftrightarrow p_H = \frac{1 + \beta_H}{2(1 + \gamma)} + \frac{\gamma}{2(1 + \gamma)}p_L - \frac{1}{2} \frac{f_H N_H + f_L N_L}{(N_H + N_L)}, \\ \frac{\partial \Pi}{\partial p_L} = 0 & \Leftrightarrow p_L = \frac{1 + \beta_L}{2} + \frac{\gamma}{2}p_H - \frac{1 + \gamma}{2} \frac{f_H N_H + f_L N_L}{(N_H + N_L)}, \\ \frac{\partial \Pi}{\partial f_H} = 0 & \Leftrightarrow f_H = \frac{1 + \theta_H}{2} - \frac{1}{2} \frac{p_H D_H + p_L D_L}{(D_H + D_L)}, \\ \frac{\partial \Pi}{\partial f_L} = 0 & \Leftrightarrow f_L = \frac{1 + \theta_L}{2} - \frac{1}{2} \frac{p_H D_H + p_L D_L}{(D_H + D_L)}.\end{aligned}$$

Define the consumer-side and seller-side average prices as

$$\bar{p} = \frac{p_H D_H + p_L D_L}{(D_H + D_L)} \quad \text{and} \quad \bar{f} = \frac{f_H N_H + f_L N_L}{(N_H + N_L)}.$$

Plugging the average prices into the above FOCs gives the following results on the interior:

$$f_L^* - f_H^* = \frac{\theta_L - \theta_H}{2},$$

$$p_H^* - p_L^* = \hat{B} + \hat{T} \bar{f}^*,$$

with $\hat{B} = B = \frac{(2-\gamma)(1+\beta_H)-(2+\gamma)(1+\beta_L)}{4+4\gamma-\gamma^2} > 0$ and $\hat{T} = T = -\frac{2\gamma}{4+4\gamma-\gamma^2} < 0$.

Supplementary Materials C: Welfare Issues

Mandatory uniform pricing. Supposing that uniform pricing is on the interior, high-type consumers' surplus, low-type consumers' surplus, sellers' surplus, and total surplus are

$$CS_H^- = \int_p^{1+\beta_H} \int_f^{1+\theta} (v_H - p) dv_S dv_H = \frac{1}{2}(1 + \theta - f)(1 + \beta_H - p)^2,$$

$$CS_L^- = \int_p^{1+\beta_L} \int_f^{1+\theta} (v_L - p) dv_S dv_L = \frac{1}{2}(1 + \theta - f)(1 + \beta_L - p)^2,$$

$$\begin{aligned} RS^- &= \int_f^{1+\theta} \int_p^{1+\beta_H} (v_S - f) dv_H dv_S + \int_f^{1+\theta} \int_p^{1+\beta_L} (v_S - f) dv_L dv_S \\ &= \frac{1}{2}[(1 + \beta_H - p) + (1 + \beta_L - p)](1 + \theta - f)^2, \end{aligned}$$

$$\begin{aligned} TW^- &= CS_H^- + CS_L^- + RS^- + \Pi^- \\ &= N \left\{ \frac{(1 + \beta_H)^2 - p^2}{2} + \frac{(1 + \beta_L)^2 - p^2}{2} + \frac{(D_H + D_L)(1 + \theta + f)}{2} \right\}. \end{aligned}$$

Price discrimination without consumers' fairness concerns. On the interior, high-type consumers' surplus, low-type consumers' surplus, sellers' surplus, and total surplus are

$$CS_H^\circ = \int_{p_H}^{1+\beta_H} \int_f^{1+\theta} (v_H - p_H) dv_S dv_H = \frac{1}{2}(1 + \theta - f)(1 + \beta_H - p_H)^2,$$

$$CS_L^\circ = \int_{p_L}^{1+\beta_L} \int_f^{1+\theta} (v_L - p_L) dv_S dv_L = \frac{1}{2}(1 + \theta - f)(1 + \beta_L - p_L)^2,$$

$$\begin{aligned} RS^\circ &= \int_f^{1+\theta} \int_{p_H}^{1+\beta_H} (v_S - f) dv_H dv_S + \int_f^{1+\theta} \int_{p_L}^{1+\beta_L} (v_S - f) dv_L dv_S \\ &= \frac{1}{2}[(1 + \beta_H - p_H) + (1 + \beta_L - p_L)](1 + \theta - f)^2, \end{aligned}$$

$$\begin{aligned} TW^\circ &= CS_H^\circ + CS_L^\circ + RS^\circ + \Pi^\circ \\ &= N \left\{ \frac{(1 + \beta_H)^2 - p_H^2}{2} + \frac{(1 + \beta_L)^2 - p_L^2}{2} + \frac{(D_H + D_L)(1 + \theta + f)}{2} \right\} \end{aligned}$$

Price discrimination with consumers' fairness concerns. On the interior, high-type consumers' surplus, low-type consumers' surplus, sellers' surplus, and total surplus are

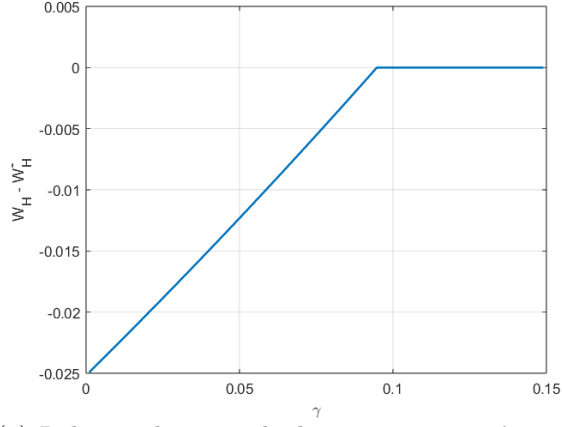
$$\begin{aligned} CS_H &= \int_{p_H + \gamma(p_H - p_L)}^{1 + \beta_H} \int_f^{1 + \theta} [v_H - p_H - \gamma(p_H - p_L)] dv_S dv_H \\ &= \frac{1}{2}(1 + \theta - f)[1 + \beta_H - p_H - \gamma(p_H - p_L)]^2, \end{aligned}$$

$$CS_L = \int_{p_L}^{1 + \beta_L} \int_f^{1 + \theta} (v_L - p_L) dv_S dv_L = \frac{1}{2}(1 + \theta - f)(1 + \beta_L - p_L)^2,$$

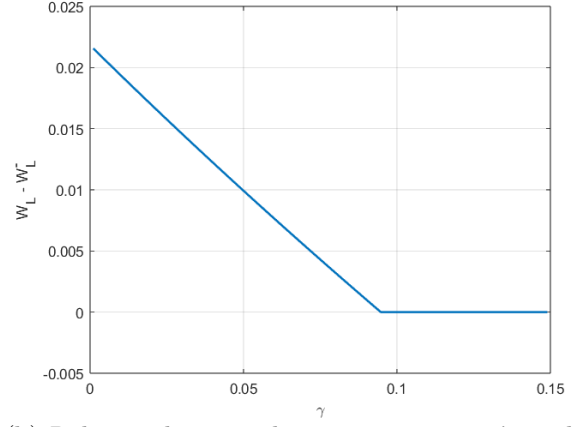
$$\begin{aligned} RS &= \int_f^{1 + \theta} \int_{p_H + \gamma(p_H - p_L)}^{1 + \beta_H} (v_S - f) dv_H dv_S + \int_f^{1 + \theta} \int_{p_L}^{1 + \beta_L} (v_S - f) dv_L dv_S \\ &= \frac{1}{2}\{[1 + \beta_H - p_H - \gamma(p_H - p_L)] + (1 + \beta_L - p_L)\}(1 + \theta - f)^2, \end{aligned}$$

$$\begin{aligned} TW &= CS_H + CS_L + RS + \Pi \\ &= N\left\{\frac{(1 + \beta_H)^2 - [p_H + \gamma(p_H - p_L)]^2}{2} + \frac{(1 + \beta_L)^2 - p_L^2}{2} + \frac{(D_H + D_L)(1 + \theta + f)}{2} - \right. \\ &\quad \left. \gamma(p_H - p_L)D_H\right\}. \end{aligned}$$

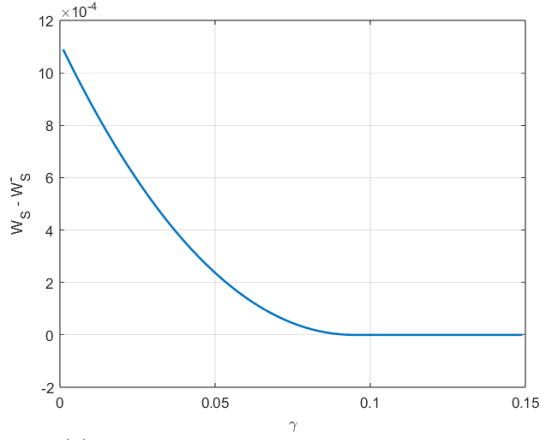
According to the total surplus expressions, a positive fairness concerns term (i.e., $\gamma(p_H - p_L) > 0$) always induces a type of “deadweight loss” to the social planner. Hence, the social optimum on the interior can be achieved only if the fairness concerns term is equal to 0. Comparing TW^- and TW° with TW , it is easy to find that $TW^- = TW^\circ > TW$ holds for price discrimination on the interior. For illustration, we present welfare simulations under price-discrimination with consumers' fairness concerns in comparison with the other two cases. The results are reported in Supplementary Figures [S3](#) and [S4](#).



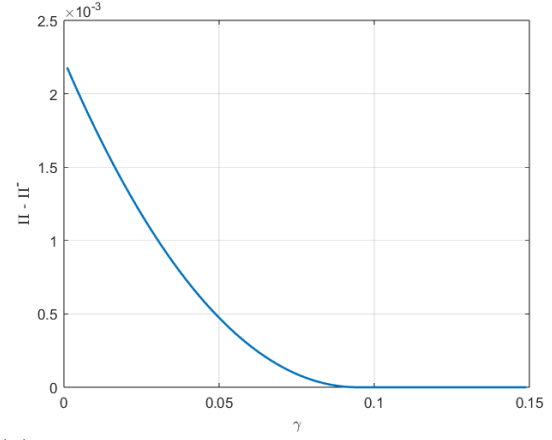
(a) Relative change in high-type consumers' surplus



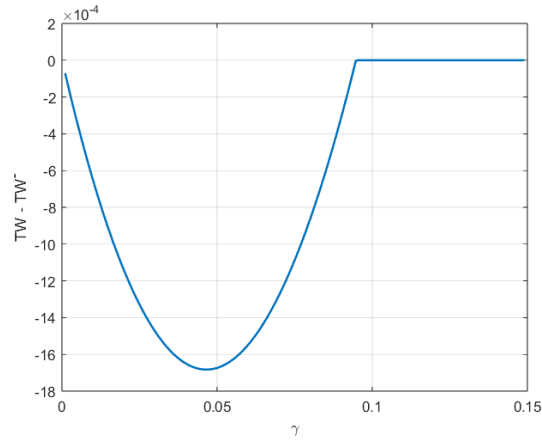
(b) Relative change in low-type consumers' surplus



(c) Relative change in sellers' surplus

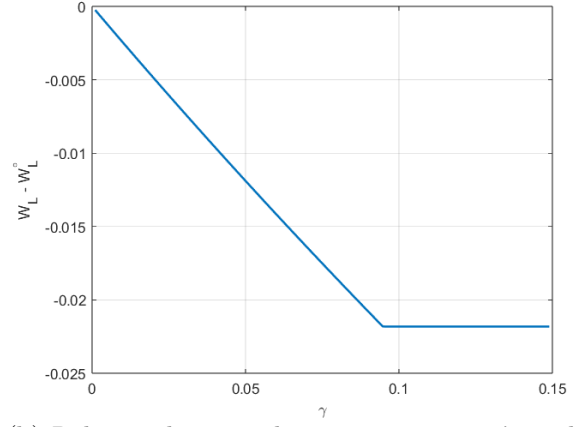
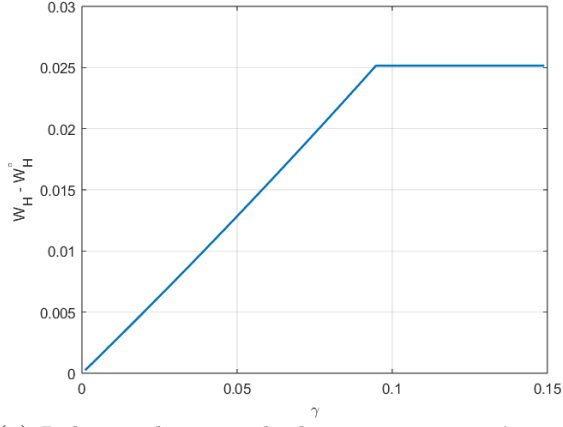


(d) Relative change in the platform's total profit

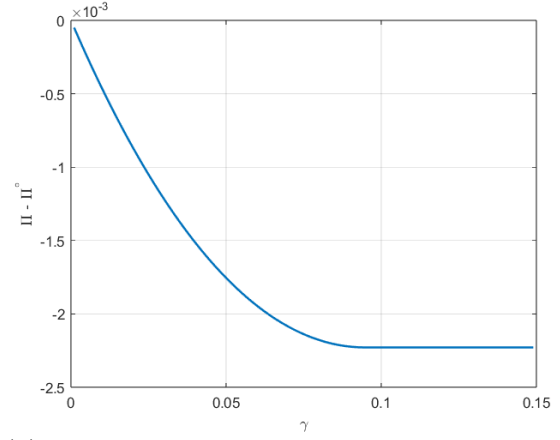
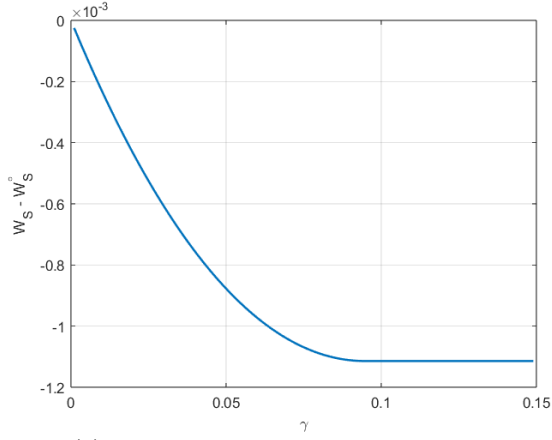


(e) Relative change in total surplus

Figure S3: Welfare comparison between the cases of price discrimination with consumers' fairness concerns and of mandatory uniform pricing. Parameter values: $\beta_H = 0.3$, $\beta_L = 0.15$, $\theta = 0.15$, and $\gamma \in (0, 0.15)$.

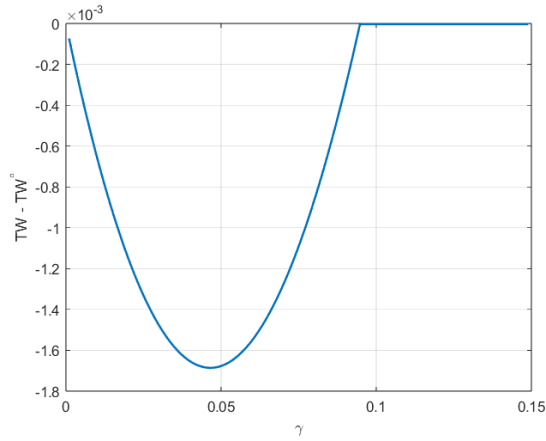


(a) Relative change in high-type consumers' surplus (b) Relative change in low-type consumers' surplus



(c) Relative change in sellers' surplus

(d) Relative change in the platform's total profit



(e) Relative change in total surplus

Figure S4: Welfare comparison between the cases with and without consumers' fairness concerns. Parameter values: $\beta_H = 0.3$, $\beta_L = 0.15$, $\theta = 0.15$, and $\gamma \in (0, 0.15)$.